SDS188 A1S2 2026 - FINAL CRAM / MARKS-RESCUE SHEET

Practice-test anchored | Excel-first | black/white/grey | learn the ROUTES, not the practice answers

EMSLearn numerical entry: 4 decimals + FULL STOP + no units/currency. **Your Excel locale:** decimal comma + semicolon separators. **Never round intermediate cells.**

FIRST 90 SECONDS - WRITE THIS ON ROUGH SHEET (IF PERMITTED)

- H1 > => RIGHT tail | H1 < => LEFT tail | H1 != => TWO tails
- sigma KNOWN => Z. sigma UNKNOWN / sample s supplied => t, df = n-1.
- CL => alpha = 1-CL. Two-tail => alpha/2 in EACH tail. One-tail => full alpha in that tail.
- DIST = value -> probability. INV = probability -> boundary. 2T = two tails.
- BETWEEN = cumulative at upper - cumulative at lower.
- CI = estimate +/- critical x SE. Power = 1-beta.
- Type I (alpha) = reject TRUE H0. Type II (beta) = do NOT reject FALSE H0.
- p-value <= alpha => reject H0. Use "reject" / "do not reject", never "accept H0".

EXCEL HELPER BLOCK - BUILD IT BEFORE LONG QUESTIONS

- Raw data: n = COUNT(range); mean = AVERAGE(range); sample s = STDEV.S(range).
- Create labelled cells: n | mean | s | sigma | alpha | df | SE | crit | test stat | p | beta | power.
- If sigma is GIVEN, use it. Do NOT replace it with STDEV.S.
- Lock ranges with $ when dragging formulas. Check COUNT before trusting any raw-data answer.
- Underline the requested OUTPUT UNIT: probability? z/t? seconds? sample size? This prevents scale mistakes.

Q1 - BOOTSTRAP CI FOR POPULATION MEAN

- Use when the population distribution is unknown / cannot reasonably be assumed normal (practice option: small sample + no idea of distribution).
- Original sample is drawn WITHOUT replacement; bootstrap resamples are drawn FROM that sample WITH replacement; every resample keeps the same n.
- For a CI for mu, calculate ONE MEAN per resample. Example row: =AVERAGE(D2:W2), then fill down.
- Order the BOOTSTRAP MEANS - not the original raw sample. 95% => 2.5% each tail.
- Use SMALL/LARGE only when the course/question tells you the cutoff position. With 250 resamples, 0.025x250=6.25, so do not invent a 6th/7th/interpolation rule if none is supplied.

Q2 - POPULATION PROPORTION CI

- p-hat = successes / n.
- SE(p-hat) = SQRT(p-hat*(1-p-hat)/n).
- CI = p-hat +/- z*SE. For 95%, z = NORM.S.INV(0.975).
- Upper limit only: p-hat + z*SE. Lower only: p-hat - z*SE.
- EXAM TRAP: if a later sub-part says "ASSUME p-hat=... and SE=...", RESET and use those new supplied values - do not carry your earlier numbers forward.

Excel: =SQRT(p*(1-p)/n) | =p+NORM.S.INV(0,975)*SE

Q3 - CRITICAL VALUE + DECISION (UNKNOWN sigma => t)

- 1) Read H1 FIRST to choose tail. 2) sample s supplied / sigma unknown => Student t. 3) df=n-1.
- RIGHT: =T.INV(1-alpha;df) | LEFT: =T.INV(alpha;df) | TWO: =T.INV.2T(alpha;df).
- Right-tail decision: tcalc > tcrit => reject H0. Left-tail: tcalc < tcrit => reject. Two-tail: ABS(tcalc) > positive tcrit => reject.
- Practice anchor: n=58 -> df=57; alpha=.05 upper-tail tcrit about 1.6720; given tcalc=2 => reject H0.

Q4 - RAW-DATA MEAN TEST, CRITICAL BOUNDARY IN ORIGINAL UNITS + POWER

- If population sigma is explicitly given => Z route. From the worksheet get n with COUNT; mean only if asked / needed.
- RIGHT-tail critical z = NORM.S.INV(1-alpha). BUT if question asks "critical value in seconds" (or original units), DO NOT submit z.
- Critical sample-mean boundary: xbar_crit = mu0 + zcrit*(sigma/SQRT(n)) for upper tail. Easiest: NORM.INV(1-alpha;mu0;sigma/SQRT(n)).
- Power at stated true mean mu1 = probability of landing in rejection region under mu1. Upper tail: 1-NORM.DIST(xbar_crit;mu1;SE;TRUE).
- beta = 1-Power. If confidence drops (e.g. 99% -> 95%), alpha rises, rejection becomes easier => beta decreases and power increases (all else fixed).

Practice anchor: mu0=10, sigma=5, n=50, alpha=.01 -> xbar_crit=11.6450; mu1=13 -> power=0.9723

DECISION / SANITY CHECKS THAT SAVE EASY MARKS

- Probability / p-value / power / beta must be between 0 and 1.
- Right-tail small alpha critical z/t should normally be POSITIVE; a tiny number like 0.063 is likely a probability, not a critical t.
- If output requested in seconds/rands/apps/etc., your final answer must be on that raw scale, not a z or t score.
- For proportions, p-hat must be between 0 and 1. Required sample size must be a whole number and is rounded UP.
- Use the question values for the current sub-part, not values you think "should" carry over.

PAGE 2 - Q5-Q8 + EXCEL FUNCTION MAP + "IF THE REAL TEST MOVES" SAFETY NET

Q5 - REQUIRED SAMPLE SIZE FOR A PROPORTION (NO PRIOR p)

- No prior/pilot estimate of population proportion => use p=0.5 (conservative maximum variance).
- n0 = z^2*p*(1-p) / e^2, where e = margin of error (use 0.06 for +/-0.06).
- ALWAYS ROUND REQUIRED n UP - never nearest, never down.
- 95% => z=NORM.S.INV(0.975).

Excel: =ROUNDUP((NORM.S.INV(0,975)^2*0,5*0,5)/(e^2);0)

Practice anchor: 95%, e=0.06 -> raw n=266.768... -> minimum n=267

Q6 - FINITE POPULATION SAMPLE SIZE (WITHOUT REPLACEMENT)

- Cue words: finite population N + sampling WITHOUT replacement => finite population adjustment.
- First calculate ordinary/infinite-population n0 using the correct mean/proportion sample-size formula.
- Then adjust: n = n0*N / (n0 + N - 1).
- Round the FINAL adjusted n UP.
- If there is no finite N / sampling is with replacement, do not force the FPC.

Excel: =ROUNDUP((n0*N)/(n0+N-1);0)

Q7 - TYPE I / TYPE II / POWER - 5-SECOND RECOGNITION

- Type I = reject H0 when H0 is TRUE. Think FALSE ALARM. Probability = alpha.
- Type II = do NOT reject H0 when H0 is FALSE. Think MISS. Probability = beta.
- Power = correctly reject a false H0 = 1-beta.
- Practice Q7 answer pattern: "not sufficient evidence" even though the true mean really IS different => Type II.
- All else fixed: alpha increases => beta decreases => power increases.

Q8 - SAMPLE-MEAN PROBABILITY WHEN sigma UNKNOWN

- Population normal + sigma unknown + sample s supplied => use t, not normal.
- SE = s/SQRT(n); df=n-1.
- Standardise BOTH boundaries: tL=(L-mu)/SE; tU=(U-mu)/SE.
- BETWEEN = T.DIST(tU;df;TRUE) - T.DIST(tL;df;TRUE).
- Practice anchor: mu=19, n=25, s=4, 18.5<Xbar<19.5 -> SE=.8, t=+/-0.625, probability=0.4621.

EXCEL FUNCTION MAP - KNOW WHAT GOES IN / COMES OUT

- NORM.DIST(x;mu;sd;TRUE) = LEFT cumulative normal probability.
- 1-NORM.DIST(...) = RIGHT normal probability. BETWEEN = upper CDF - lower CDF.
- NORM.S.INV(prob) = probability -> z-score.
- NORM.INV(prob;mu;sd) = probability -> RAW boundary in original units.
- T.DIST(t;df;TRUE) = LEFT cumulative t probability.
- T.DIST.2T(ABS(t);df) = TWO-tail t probability.
- T.INV(prob;df) = left-cumulative probability -> t boundary.
- T.INV.2T(alpha;df) = positive TWO-tail critical t.
- COUNT / AVERAGE / STDEV.S / SQRT / ROUNDUP / SMALL / LARGE = your core execution tools.

IF THE REAL TEST MOVES BEYOND THE PRACTICE - CH6/7 SAFETY STRIP

- Normal X: LEFT=NORM.DIST(x;mu;sigma;TRUE); RIGHT=1-left; BETWEEN=CDF(U)-CDF(L); inverse=NORM.INV(left_area;mu;sigma).
- Sample mean with sigma known: same normal logic BUT sd becomes SE=sigma/SQRT(n). Never use sigma directly for Xbar.
- Central C interval for Xbar: lower=NORM.INV((1-C)/2;mu;SE); upper=NORM.INV((1+C)/2;mu;SE).
- Uniform(a,b): P(c<X<d)=(d-c)/(b-a); mean=(a+b)/2; SD=(b-a)/SQRT(12).
- Exponential: match time units with lambda. LEFT=EXPON.DIST(x;lambda;TRUE); RIGHT=1-left.
- Normal approximation to binomial: check n*pi>5 AND n*(1-pi)>5; mean=n*pi; SD=SQRT(n*pi*(1-pi)); continuity correction may be required.

MARKS-RESCUE HACKS - USE THESE WHEN YOUR MIND GOES BLANK

- 1) Translate WORDS before numbers: greater/right, less/left, different/two-tail, unknown sigma/t.
- 2) Ask "What does the answer box want?" Probability, critical statistic, raw boundary, CI limit, n, or decision?
- 3) Use helper cells and references instead of one giant formula. Keep full precision until final entry.
- 4) If you get a weird answer, check scale: probability 0-1; proportion 0-1; sample size integer; df=n-1; original-unit boundary should look like the original variable.
- 5) If a later part supplies new values, use them. This exact trap appears in Practice Q2.
- 6) For raw data, COUNT first. Missing/blank cells can silently change n.
- 7) Read sigma vs s carefully: given population sigma => Z; only sample s => t.
- 8) Panic route: TARGET -> TAIL -> Z/t -> SE -> DIST/INV -> Excel -> units/range check -> EMSLearn 4 decimals with FULL STOP.